CD Lab 5

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Ast.h

ast.h
~/sem6/CD/PE-5

```
#ifndef AST_H
#define AST_H

typedef struct Node {
    char *value;
    struct Node *left;
    struct Node *right;
} Node;

Node* createNode(char *val, Node *l, Node *r);
void postorder(Node *root);
void freeTree(Node *root);

#endif
```

C/ObjC Header Tab Width: 8 Ln 14, Col 7 INS

Ast.c

ast.c
~/sem6/CD/PE-5

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "ast.h"

/* extern file pointer from parser */
extern FILE *outfile;

Node* createNode(char *val, Node *l, Node *r) {
    Node *node = (Node*)malloc(sizeof(Node));
    node->value = strdup(val);
    node->left = l;
    node->right = r;
    return node;
}

void postorder(Node *root) {
    if (root == NULL) return;

    postorder(root->left);
    postorder(root->right);
    fprintf(outfile, "%s ", root->value);  // ✅ write to file
}

void freeTree(Node *root) {
    if (root == NULL) return;

    freeTree(root->left);
    freeTree(root->right);
    free(root->value);
    free(root);
}
```

Bracket match found on line: 25 C Tab Width: 8 Ln 32, Col 2 INS

Lexer.l

Text Editor

Open | lexer.l ~/sem6/CD/PE-5 | Save

```
%option noyywrap

%{
#include "ast.h"
#include "parser.tab.h"
#include <stdlib.h>
#include <string.h>
%}

%%

[0-9]+ {
    yylval.str = strdup(yytext);
    return NUM;
}

[ \t]+    ;          /* ignore whitespace */

\n        return '\n';

"+"       return '+';
"-"       return '-';
"*"       return '*';
"/"       return '/';
"("       return '(';
")"       return ')';

. {
    printf("Invalid character: %s\n", yytext);
}

%%
```

Lex | Tab Width: 8 | Ln 32, Col 3 | INS

Parser.y

Open

parser.y
~/sem6/CD/PE-5

Save

```
%{
#include <stdio.h>
#include <stdlib.h>
#include "ast.h"

extern int yylex();
extern FILE *yyin;

void yyerror(const char *s);

Node *root;
FILE *outfile;
%}

%code requires {
    #include "ast.h"
}

%union {
    Node* node;
    char* str;
}

%token <str> NUM
%type <node> expr

%left '+' '-'
%left '*' '/'

%%

input:
      /* empty */
    | input line
;

line:
```

Yacc Tab Width: 8 Ln 78, Col 2 INS

parser.y
~/sem6/CD/PE-5

```
            freeTree(root);
        }
    | '\n'
;

expr:
        expr '+' expr { $$ = createNode("+", $1, $3); }
    | expr '-' expr { $$ = createNode("-", $1, $3); }
    | expr '*' expr { $$ = createNode("*", $1, $3); }
    | expr '/' expr { $$ = createNode("/", $1, $3); }
    | '(' expr ')'  { $$ = $2; }
    | NUM           { $$ = createNode($1, NULL, NULL); }
;

%%

void yyerror(const char *s) {
    fprintf(stderr, "Error: %s\n", s);
}

int main() {
    yyin = fopen("input.txt", "r");
    outfile = fopen("output.txt", "w");

    if (!yyin || !outfile) {
        printf("Error opening file\n");
        return 1;
    }

    yyparse();

    fclose(yyin);
    fclose(outfile);

    printf("Output written to output.txt\n");
    return 0;
}
```

Yacc Tab Width: 8 Ln 78, Col 2 INS

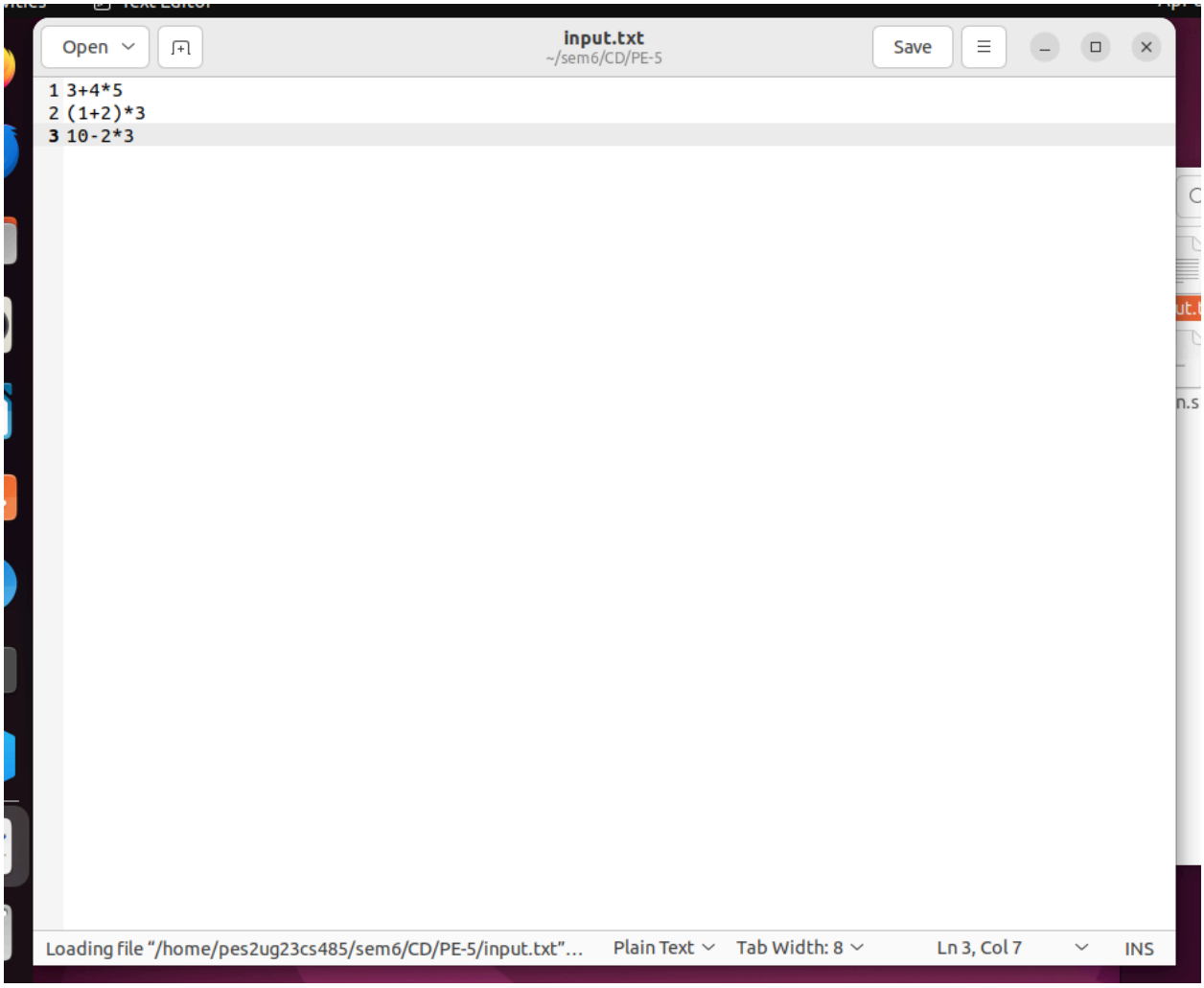
[run.sh]

run.sh
~/sem6/CD/PE-5

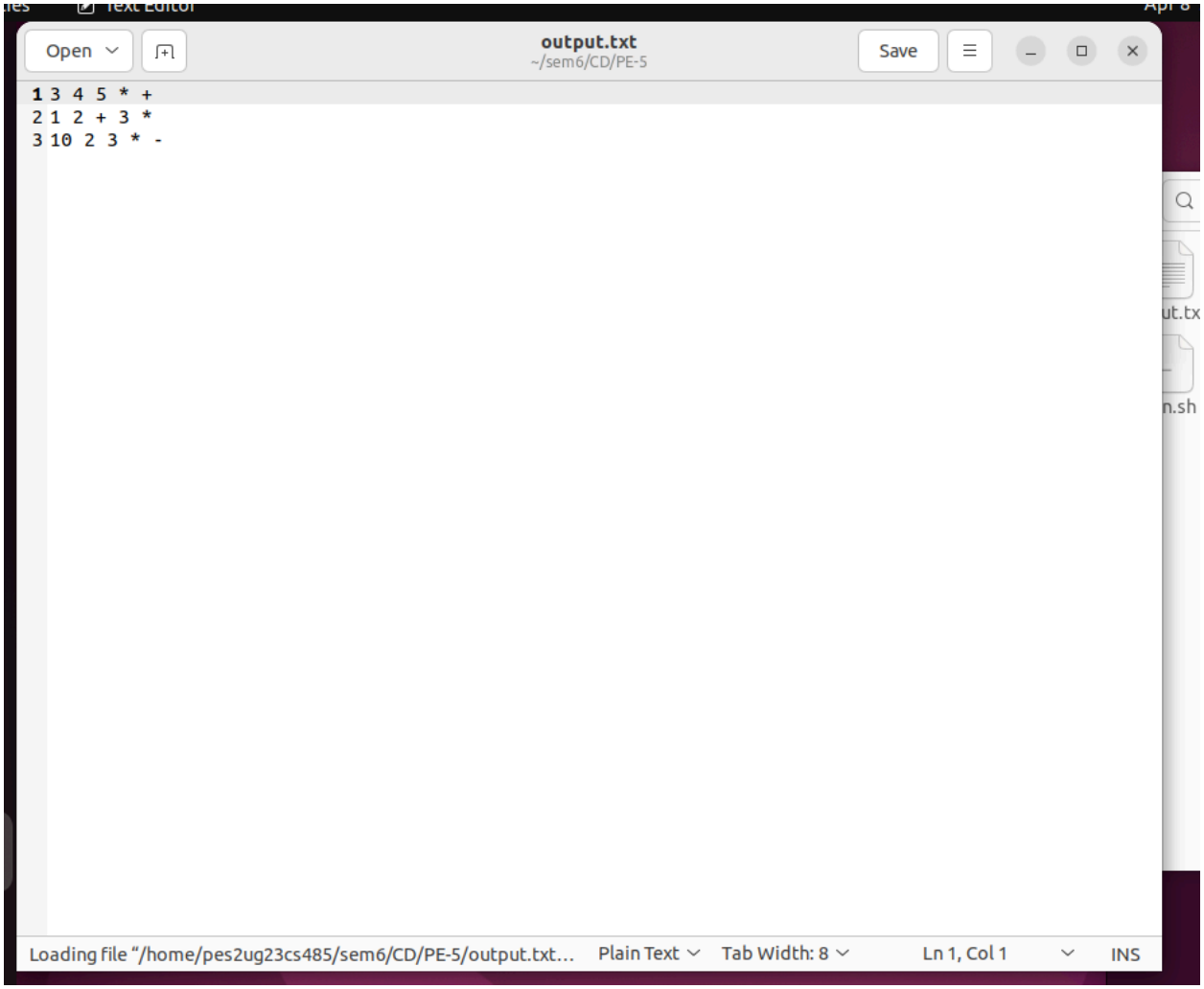
```bash
#!/bin/bash

echo "Cleaning old files..."
rm -f parser.tab.* lex.yy.c ast_parser output.txt

echo "Running Bison..."
bison -d parser.y

echo "Running Flex..."
flex lexer.l

echo "Compiling..."
gcc parser.tab.c lex.yy.c ast.c -o ast_parser

echo "Executing..."
./ast_parser

echo "Done! Check output.txt"
```

Loading file "/home/pes2ug23cs485/sem6/CD/PE-5/run.sh"... sh Tab Width: 8 Ln 18, Col 30 INS

Input


Open
input.txt
~/sem6/CD/PE-5
Save
1 3+4*5
2 (1+2)*3
3 10-2*3
Loading file "/home/pes2ug23cs485/sem6/CD/PE-5/input.txt"...
Plain Text
Tab Width: 8
Ln 3, Col 7
INS

Output


Open
output.txt
~/sem6/CD/PE-5
Save
3 4 5 * +
1 2 + 3 *
10 2 3 * -
Loading file "/home/pes2ug23cs485/sem6/CD/PE-5/output.txt...
Plain Text
Tab Width: 8
Ln 1, Col 1
INS